Properties of Ionic Compounds

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Written by: Caroline Carroll
Reviewed by: Stewart Hird
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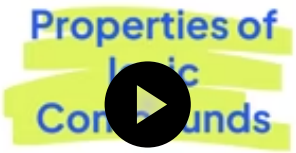

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Help me understand

Properties of ionic compounds

- Ionic compounds are usually **solid** at room temperature
- They have **high** melting and boiling points
- Ionic compounds are good conductors of electricity in the **molten** state or in **solution**
- They are poor conductors in the solid state

Explaining the properties of ionic compounds

Extended tier only

- Ionic compounds have **high melting** and **boiling points**
  - They have giant structures
  - There are strong electrostatic forces of attraction between oppositely charged ions in all directions
  - The forces need lots of energy to overcome them
- Ionic compounds are **poor conductors in the solid state**
  - The ions are in fixed positions in the lattice
  - They are therefore unable to move and carry a charge

Stuck on something?

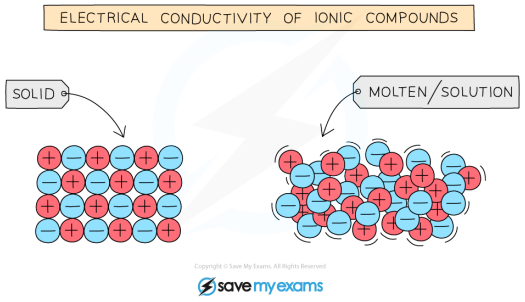
Ask questions and get help with exactly what you're revising.

Got it

water, the ions are able to move and carry a charge

- The greater the charge on the ions, the stronger the electrostatic forces and the higher the melting point will be
  - For example, magnesium oxide consists of Mg^{2+} and O^{2-} so will have a higher melting point than sodium chloride which contains the ions, Na^{+} and Cl^{-}

Diagram to show the electrical conductivity of ionic compounds



Molten or aqueous ions move freely but cannot in solid form

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